

Title for your proposal

1 Driving Motivation

Purpose of this section: Clearly articulate the compelling research problem. This section should make a program officer and panelist care about the work within the first page. If the motivation is weak, the proposal will not be competitive.

Read the NSF PAPPG carefully regarding the Project Description requirements and merit review criteria. Make sure you understand the NSF page limit requirement. This section should establish:

- The core problem or challenge.
- Why it matters scientifically, societally, or technologically.
- The gap in current knowledge, capability, or theory.
- Why now is the right time to address this problem.

You must:

- Clearly define the research question(s).
- Articulate testable hypotheses or research objectives.
- Identify the intellectual domain(s) involved.

Do not provide a literature review here. Focus on urgency, significance, and opportunity. By the end of this section, a reviewer should understand

What is the problem? Why is it important? What will change if this succeeds?

2 Prior Work

Purpose of this section: Situate your prior work [1] within the existing literature [2] and clearly demonstrate the gap your work fills. Guidance is as follows:

- Summarize the most relevant foundational and recent work.
- Identify limitations of existing approaches.
- Clarify how your approach differs.

This is not a historical survey. It is a strategic positioning section. Explicitly address:

- What has already been done?
- Why is it insufficient?
- What conceptual, methodological, or empirical gap remains?

Where appropriate, include discussion of:

- Preliminary results (if any).
- Feasibility evidence.
- Foundational theory supporting your approach.

Your writing should make clear that you understand the field deeply enough to lead it.

3 Intellectual Merit

This section is required by NSF and must explicitly address the Intellectual Merit review criterion as described in the PAPPG.

You must directly respond to:

- What is the potential for the proposed activity to advance knowledge?
- How is the work creative, original, or transformative?
- What is the quality of the research plan?
- What are the qualifications of the investigator (you)?

This section should include:

- Theoretical contributions.
- Methodological innovations.
- Expected scholarly outputs.
- Broader implications within the academic discipline.

Avoid vague claims. Provide concrete statements such as:

- “This work will establish?”
- “This research will resolve?”
- “This framework enables?”

Panelists score this explicitly. Write accordingly.

4 Broader Impacts

This section is required by NSF and must explicitly address the Broader Impacts review criterion as defined in the PAPPG. You must address:

- How does this benefit society?
- How does it contribute to desired societal outcomes?
- How will knowledge be disseminated?

Possible areas to consider (see PAPPG for full list):

- Education and workforce development
- Broadening participation
- Public engagement
- Open science and data sharing
- Partnerships with industry or community stakeholders

Broader Impacts should be integrated and realistic - not symbolic. Avoid generic statements. Provide:

- Specific activities
- Measurable outcomes
- Clear audiences

5 Research Objectives

This section should clearly state the central engineering objective that defines the intellectual direction of the project. Begin by articulating the overarching objective of the work in a single, declarative sentence that describes the primary engineering goal and the capability, system, framework, or understanding the project seeks to establish. This statement should define the unifying technical vision for the proposal.

Following the overarching objective, provide three concise numbered objectives that collectively achieve this goal. These objectives should reflect the core technical thrusts of the work (e.g., build, test, analyze), and together define a coherent and executable research agenda.

- **Objective 1: [Build / Design / Develop ...]** - One sentence describing the first technical objective, such as constructing a model, designing a system, developing an algorithm, or creating a prototype.
- **Objective 2: [Test / Validate / Analyze ...]** - One sentence describing the second technical objective, such as experimentally validating performance, benchmarking against the state of the art, or analyzing the governing mechanisms and key performance limitations.
- **Objective 3: [Explore / Generalize / Pioneer ...]** - One sentence describing the third technical objective, such as exploring new research directions, generalizing the resulting insights beyond the initial system or operating conditions, or pioneering a new capability, framework, or application.

Each objective must correspond directly to one of the research tasks described in the following section (Task 1 addresses Objective 1, Task 2 addresses Objective 2, and Task 3 addresses Objective 3). The alignment between objectives and tasks should be immediately clear to a reviewer without additional explanation. This section is typically concise and should fit naturally within the overall 15-page Project Description limit.

6 Research Plan

6.1 Overview

Provide a concise overview of the research strategy and how the tasks integrate to achieve the overall objective. The structure below is highly recommended. Each task must be technically substantive and logically connected.

6.2 Task 1: [Insert Descriptive Title]

The goal of Task 1 is to meet Objective 1. Add a basic sentence here to describe the task.

Subtask 1.1: [Title]

Add the textual description of the subtask here.

Subtask 1.2: [Title]

Add the textual description of the subtask here.

Subtask 1.3: [Title]

Add the textual description of the subtask here.

6.3 Task 2: [Insert Descriptive Title]

The goal of Task 1 is to meet Objective 2. Add a basic sentence here to describe the task.

Subtask 2.1 [Title]

Add the textual description of the subtask here.

Subtask 2.2 [Title]

Add the textual description of the subtask here.

Subtask 2.3 [Title]

Add the textural description of the subtask here.

6.4 Task 3: [Insert Descriptive Title]

The goal of Task 1 is to meet Objective 3. Add a basic sentence here to describe the task.

Subtask 3.1 [Title]

Add the textural description of the subtask here.

Subtask 3.2 [Title]

Add the textural description of the subtask here.

Subtask 3.3 [Title]

Add the textural description of the subtask here.

7 Project Management Plan

Although this is a single-investigator-style proposal, you must demonstrate:

- Timeline (include a Gantt chart if space permits).
- Milestones.
- Risk management.
- Data management alignment with NSF Data Management Plan requirements.

Discuss:

- Year-by-year objectives.
- Publication strategy.
- Conference dissemination.
- Reproducibility practices.

Your management plan should make reviewers confident the project will finish on time and produce meaningful outcomes.

8 Prior NSF Support

This section must comply exactly with the current NSF PAPPG requirements. If the PI has received NSF funding within the past five years, a “Results from Prior NSF Support” section is required. Important guidance from the PAPPG include:

- If the PI has received more than one NSF award (excluding amendments), only the award most closely related to the proposed project must be reported.
- “Support” includes salary support and any other funding awarded by NSF, including research grants, Graduate Research Fellowships (GRFP), Major Research Instrumentation awards, conference awards, equipment awards, travel awards, center awards, or similar mechanisms.

For the award being reported, include:

- NSF award number

- Award amount
- Period of support
- Summary of results
- Publications and products
- Explicit description of outcomes related to Intellectual Merit
- Explicit description of outcomes related to Broader Impacts

Being paid as a graduate student, postdoc, or staff member on an NSF award where your advisor or supervisor is the PI does not constitute prior NSF support for purposes of this section. Only awards made directly to you as PI or co-PI (including GRFP) must be reported.

If you have not received prior NSF support as defined above, state: “The PI has not received prior NSF support.” Failure to comply with PAPPG formatting and content requirements may result in administrative return without review.

References Cited

- [1] A. Downey, "Machine learning for engineering problem solving: A practical example-driven guide to classical techniques," Jul. 2025, [Online]. Available: <https://doi.org/10.31224/4909>
- [2] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "Imagenet classification with deep convolutional neural networks," *Advances in neural information processing systems*, vol. 25, 2012.